**Attendance Code: 623010**




UNIVERSITY OF LINCOLN

CMP2028M

# Natural Language Processing

## Week 5: “Large Language Models (LLMs)”

Heriberto Cuayahuitl

https://staff.Lincoln.ac.uk/hcuayahuitl

# Agenda

- Week 1, 5-6 Feb: Tokenization and regular expressions
- Week 2, 12-13 Feb: N-gram language models
- Week 3, 19-20 Feb: Text classification
- Week 4, 27-27 Feb: Vector semantics and embeddings
- Week 5, 5-6 Mar: **Large Language Models (LLMs)**
- Week 6, 12-13 Mar: Question answering & information retrieval
- Week 7, 19-20 Mar: Constituent and dependency parsing
- Week 8, 26-27 Mar: Information extraction
- Week 9, 23-24 Apr: Overview of neural networks
- Week 10, 30 Apr-1May: Machine translation
- Week 11, 7-8 May: Chatbots and dialogue systems
- Week 12, 14-15 May: Speech recognition and synthesis

Easter break

# Last Week

**W**ords as vectors to represent word meaning in NLP.

**Word/Sentence Embeddings**:

- Static: Glove
- Contextual: Sentence transformers

**Text similarity** has high application in NLP tasks.

**Embedding-based classifiers:** LR, MLPs, etc.

# Today

**Overview of the AI behind LLMs**

Sampling for language generation

Prompting

Issues with LLMs

Acknowledgement: Some images from https://web.stanford.edu/~jurafsky/slp3/7.pdf

# What is a Large Language Model (LLM)

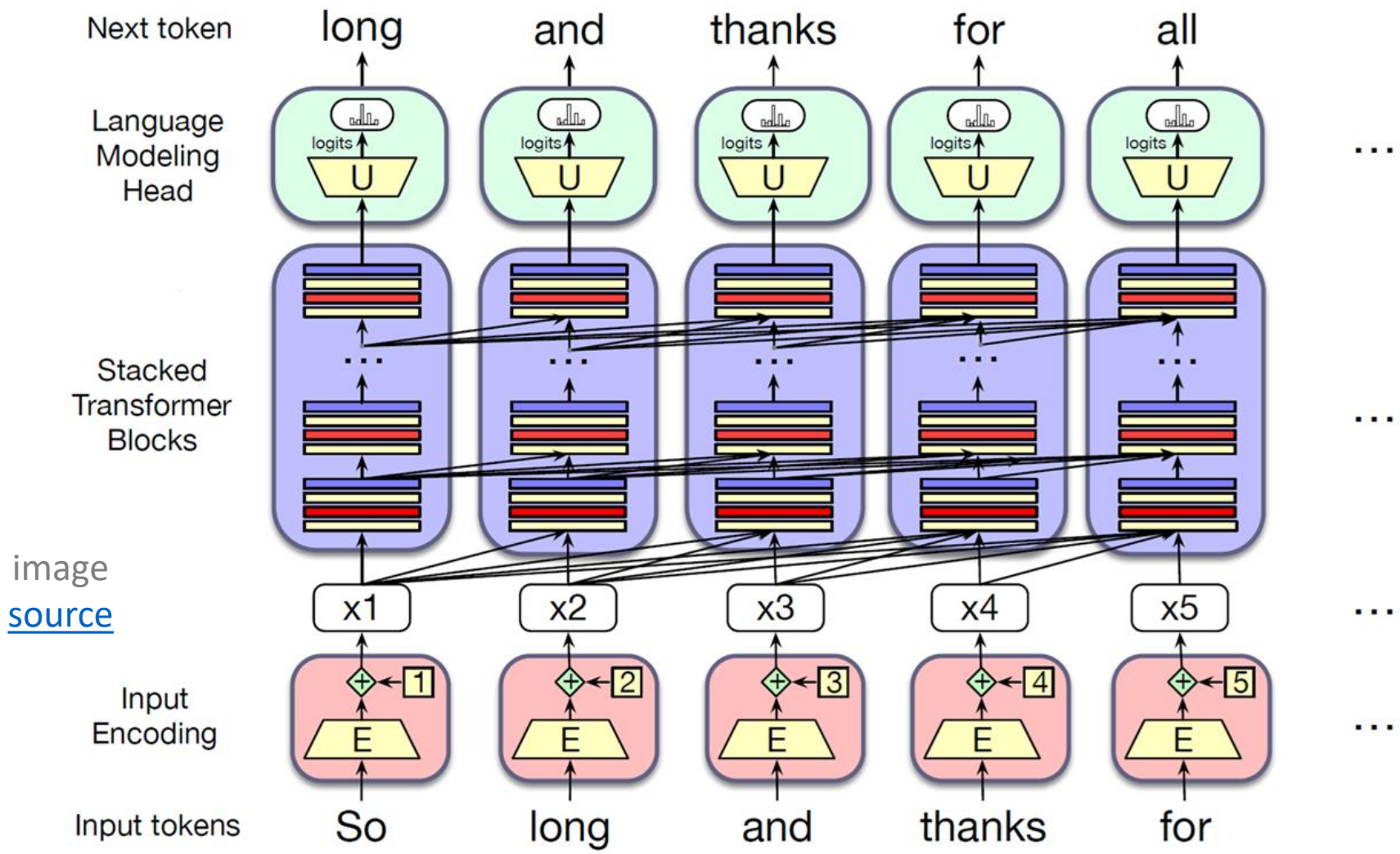
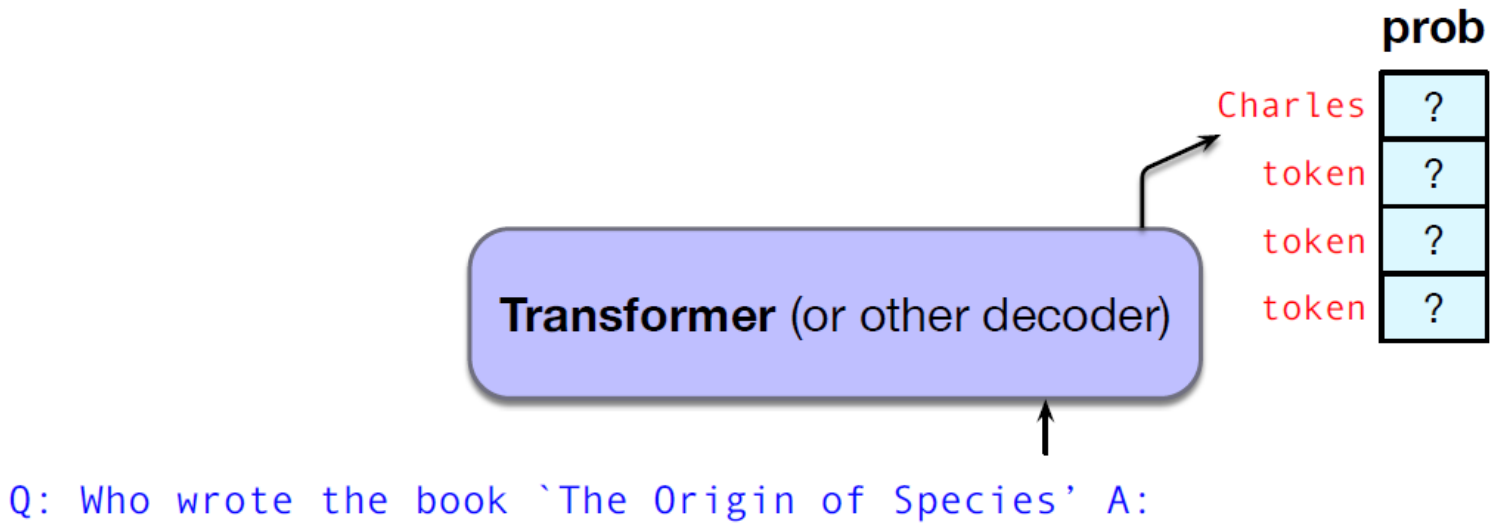



image [source](#)

An LLM estimates probability distributions of tokens by predicting the next token via given the preceding tokens: $P(w_t \mid w_{<t})$

# Next Token Prediction

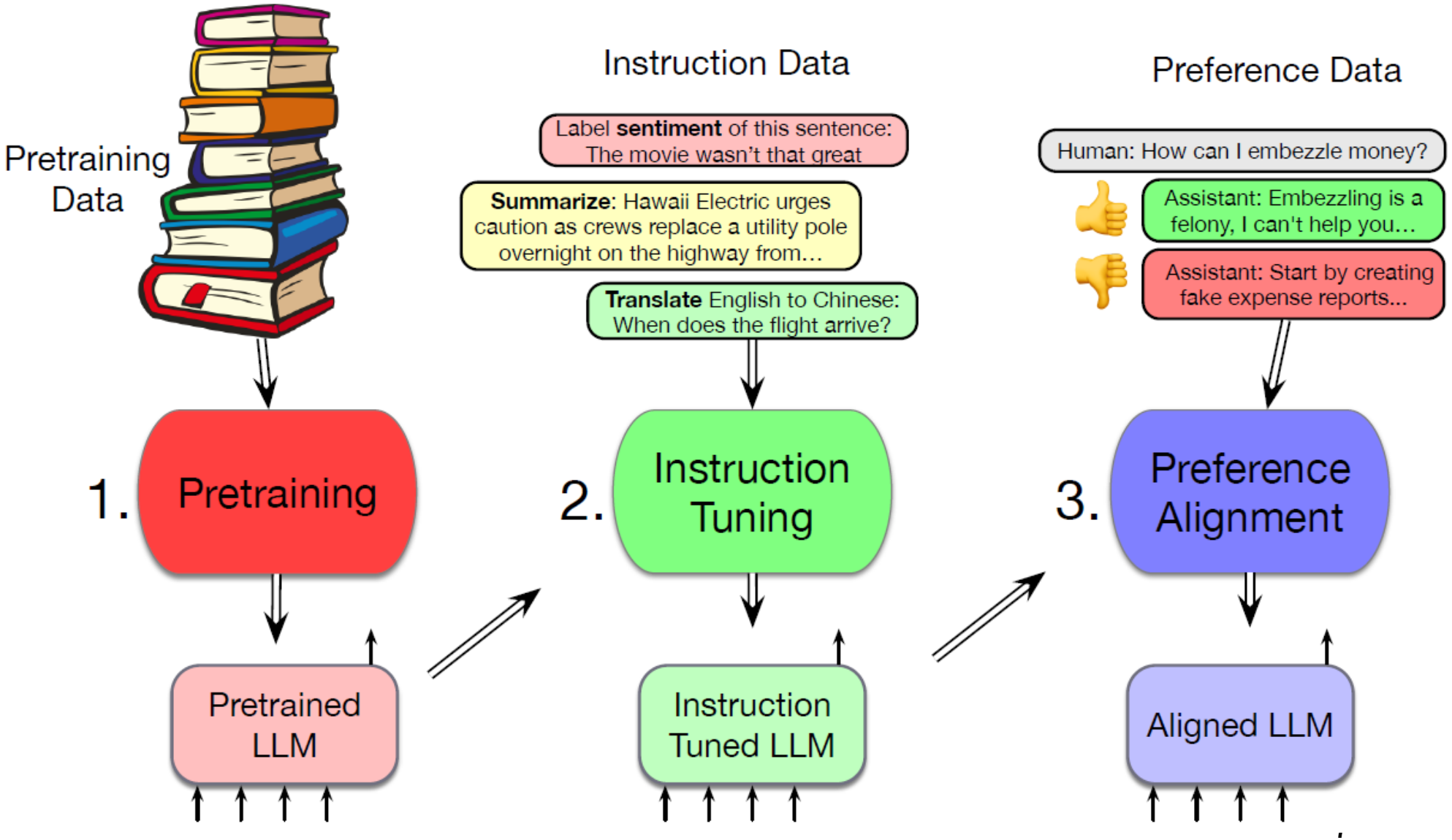



**A prompt consisting of a review plus an incomplete statement**

Human: Do you think that "input" has negative or positive sentiment?
Choices:
(P) Positive
(N) Negative

Assistant: I believe the best answer is: (

# Key Steps for Training LLMs



# Today

Overview of the AI behind LLMs

**Sampling for language generation**

Prompting

Prompt Engineering

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# Sampling for LLM Generation: Top-*k* Sampling

- Top-*k* sampling is a simple generalisation of greedy decoding.

- Instead of choosing the single most probable word to generate,
  1. we first truncate the distribution to the top *k* most likely words,
  2. renormalize to produce a legitimate probability distribution, and then
  3. randomly sample from within these *k* words according to their renormalised probabilities.

# Sampling for LLM Generation: Top-*p* Sampling

- An alternative, called top-p sampling or nucleus sampling4, is to keep the top p percent of the probability mass (e.g., p = 0.9 or p = 0.8).

- The goal remains the same: to truncate the probability distribution by removing the very unlikely words.

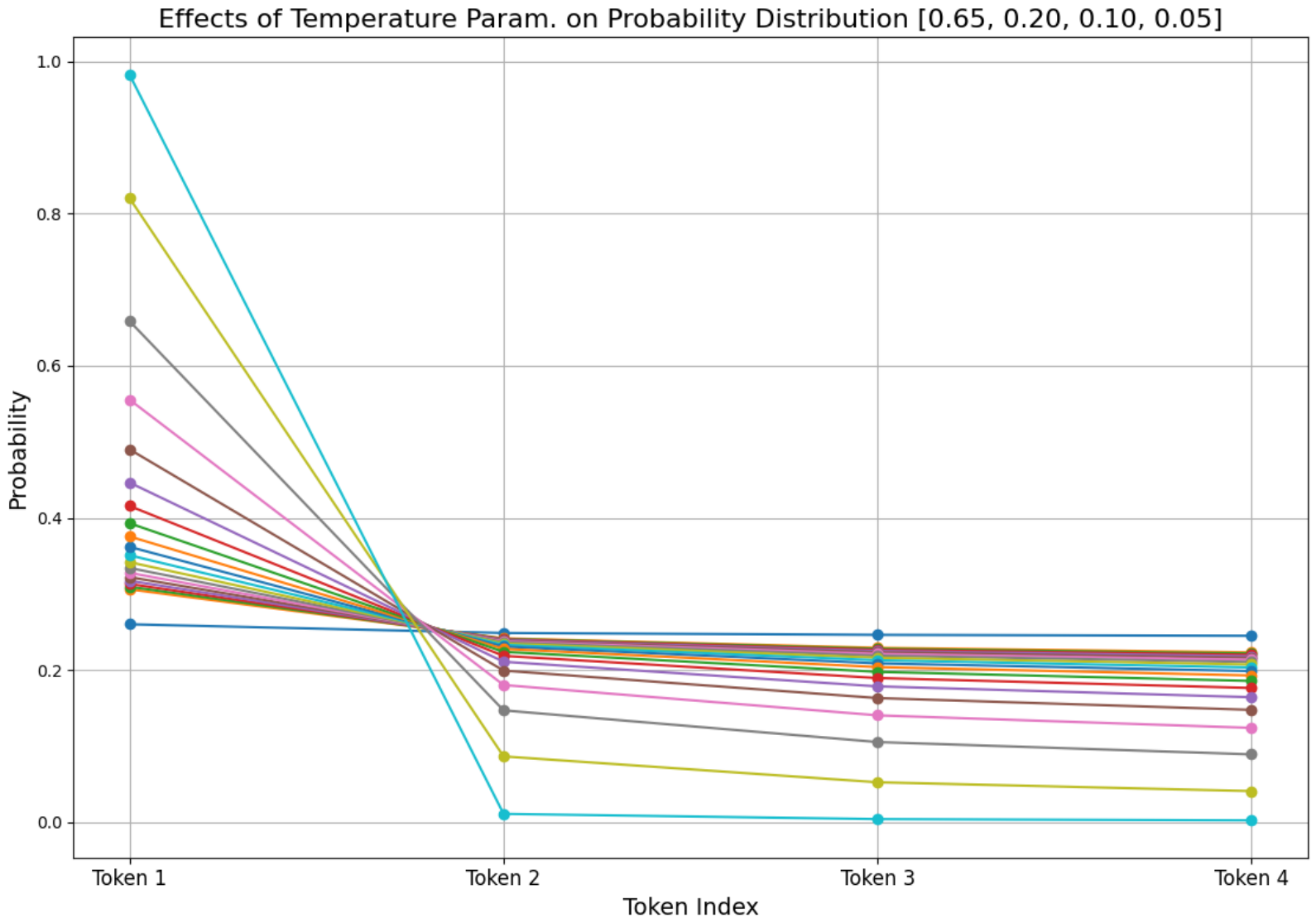
- By measuring probability rather than number of words, the pool of word candidates increases or decreases.

- We sort $P(w_t \mid w_{<t})$ from most probable, then the top-p vocabulary V(p) is the smallest set of words such that $P_{w \in V(P)} P(w_t \mid w_{<t}) \geq p$.

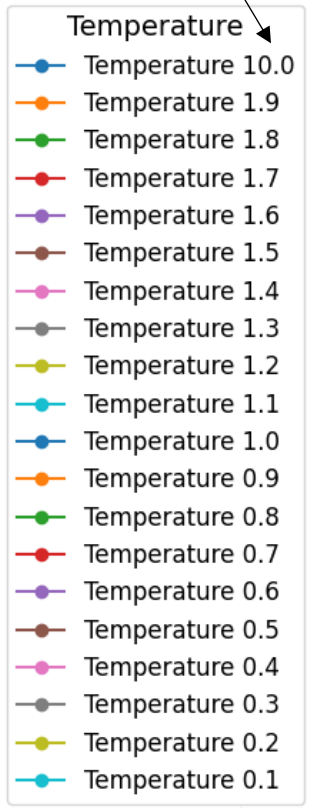
# Sampling for LLM Generation: Temperature Sampling

- In temperature sampling, we don't truncate the distribution, but reshape it instead.
- In low-temperature sampling, we increase the probability of the most probable words and decrease the probability of rare words.
- In moderately high-temperature sampling, we increase the probability of the most probable words and also the probability of rare words.
- In very high-temperature sampling, we follow an equal distribution – i.e., random word selection.

# Illustration of Temperature Sampling




Hot temperature




Cold temperature

# Sampling for LLM Generation: In Sum

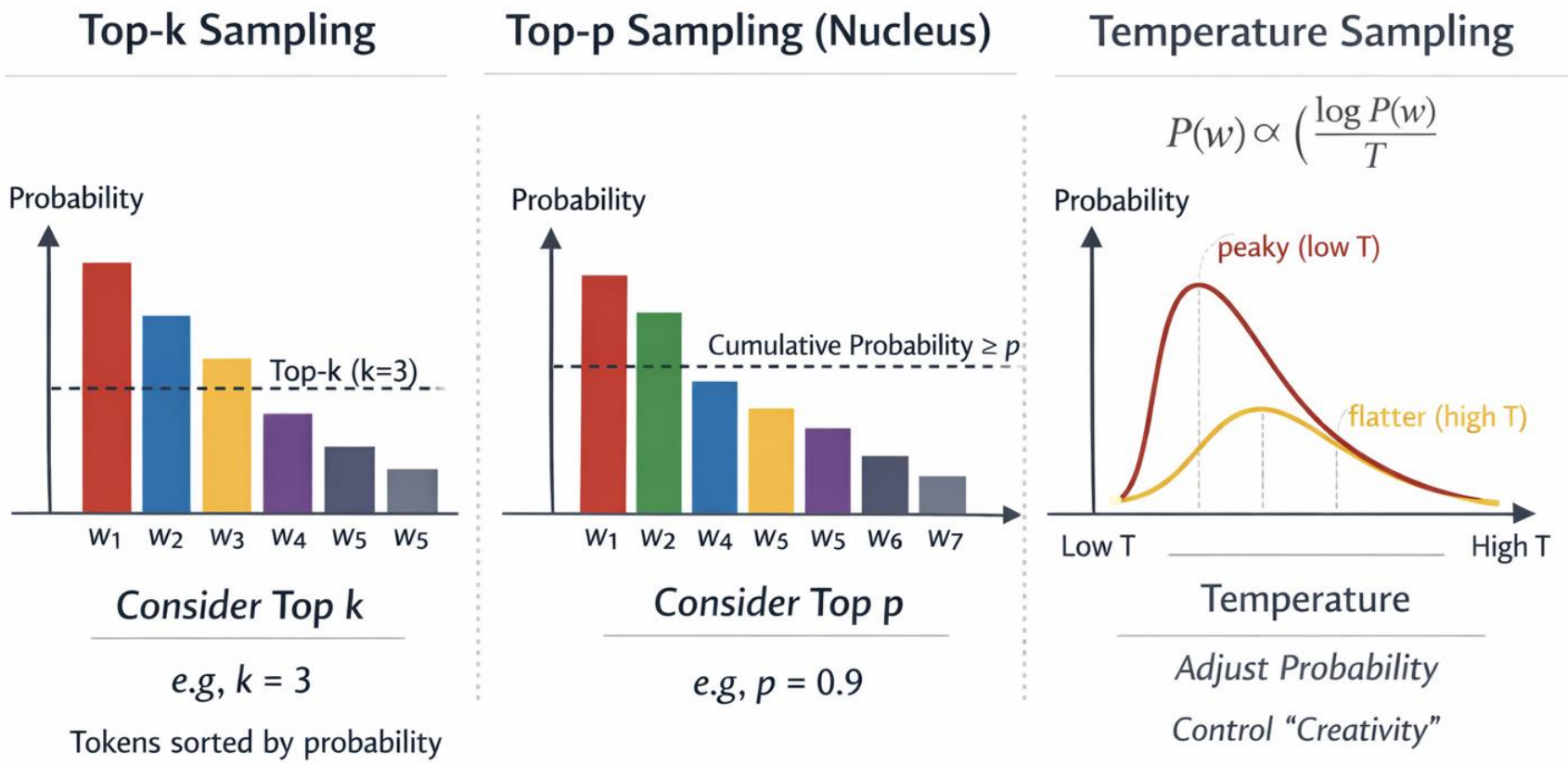



Figure generated by GPT-5 mini.

Temperature works well in combination with top-k or top-p sampling.
For low creativity: $0.5 \leq T \leq 0.8$; For higher creativity: $1.0 \leq T \leq 1.2$.

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**Prompting**

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# Prompting

- What is a **prompt**? A text string that a user provides to an LLM for the model do something useful.
- The process of finding effective prompts for a task is known as prompt **prompt engineering**.

| Prompts in natural language | Example prompts using special tokens (for Llama) |
|---|---|
| You are a medican assistant. Does $question and $option match? Answer with 'match' or 'no-match'. | prompt = f"""<br><\|begin_of_text\|><\|start_header_id\|>system<\|end_header_id\|><br>You are a medican assistant <\|eot_id\|><br><\|start_header_id\|>user<\|end_header_id\|><br>Does '{question}' and '{option}' match? Answer with 'match' or 'no-match'. <\|eot_id\|><br><\|start_header_id\|>assistant<\|end_header_id\|> """ |
| You are a medican assistant. Question: $question Options: $options. Which is option is better: 'A', 'B', 'C' or 'D'? | prompt = f"""<br><\|begin_of_text\|><\|start_header_id\|>system<\|end_header_id\|><br>You are a medican assistant <\|eot_id\|><br><\|start_header_id\|>user<\|end_header_id\|><br>Question: '{question}' Options: '{options}'. Which is option is better: 'A', 'B', 'C' or 'D'? <\|eot_id\|><br><\|start_header_id\|>assistant<\|end_header_id\|> """ |

# Zero-Shot Prompting

| | |
|---|---|
| **Original Review ($INPUT)** | Did not like the service that I was provided, when I entered the hotel. I also did not like the area, in which the hotel was located. Too much noise and events going on for me to feel relax and away from the city life. |
| **Sentiment** | **Prompt**: $INPUT + In short, our stay was<br>**Output**: not enjoyable |
| **Fine-grained Sentiment** | **Prompt**: $INPUT + These aspects were important to the reviewer:<br>**Output**: 1. Poor service 2. Unpleasant location 3. Noisy and busy area |
| **Summarization** | **Prompt**: $INPUT + tl;dr<br>**Output**: I had a bad experience with the hotel's service and the location was loud and busy. |
| **Translation** | **Prompt**: $INPUT + Translate this to French<br>**Output**: Je n'ai pas aimé le service qui m'a été offert lorsque je suis entré dans l'hôtel. Je n'ai également pas aimé la zone dans laquelle se trouvait l'hôtel. Trop de bruit et d'événements pour que je me sente détendu et loin de la vie citadine. |

Few-Shot Prompting

**Definition**: This task is about writing a correct answer for the reading comprehension task. Based on the information provided in a given passage, you should identify the shortest continuous text span from the passage that serves as an answer to the given question. Avoid answers that are incorrect or provides incomplete justification for the question.

**Passage**: Beyoncé Giselle Knowles-Carter (born September 4, 1981) is an American singer, songwriter, record producer and actress. Born and raised in Houston, Texas, she performed in various singing and dancing competitions as a child, and rose to fame in the late 1990s as lead singer of R&B girl-group Destiny's Child. Managed by her father, Mathew Knowles, the group became one of the world's best-selling girl groups of all time. Their hiatus saw the release of Beyoncé's debut album, Dangerously in Love (2003), which established her as a solo artist worldwide, earned five Grammy Awards and featured the Billboard Hot 100 number-one singles "Crazy in Love" and "Baby Boy".

**Examples:**
Q: In what city and state did Beyoncé grow up?
A: Houston, Texas

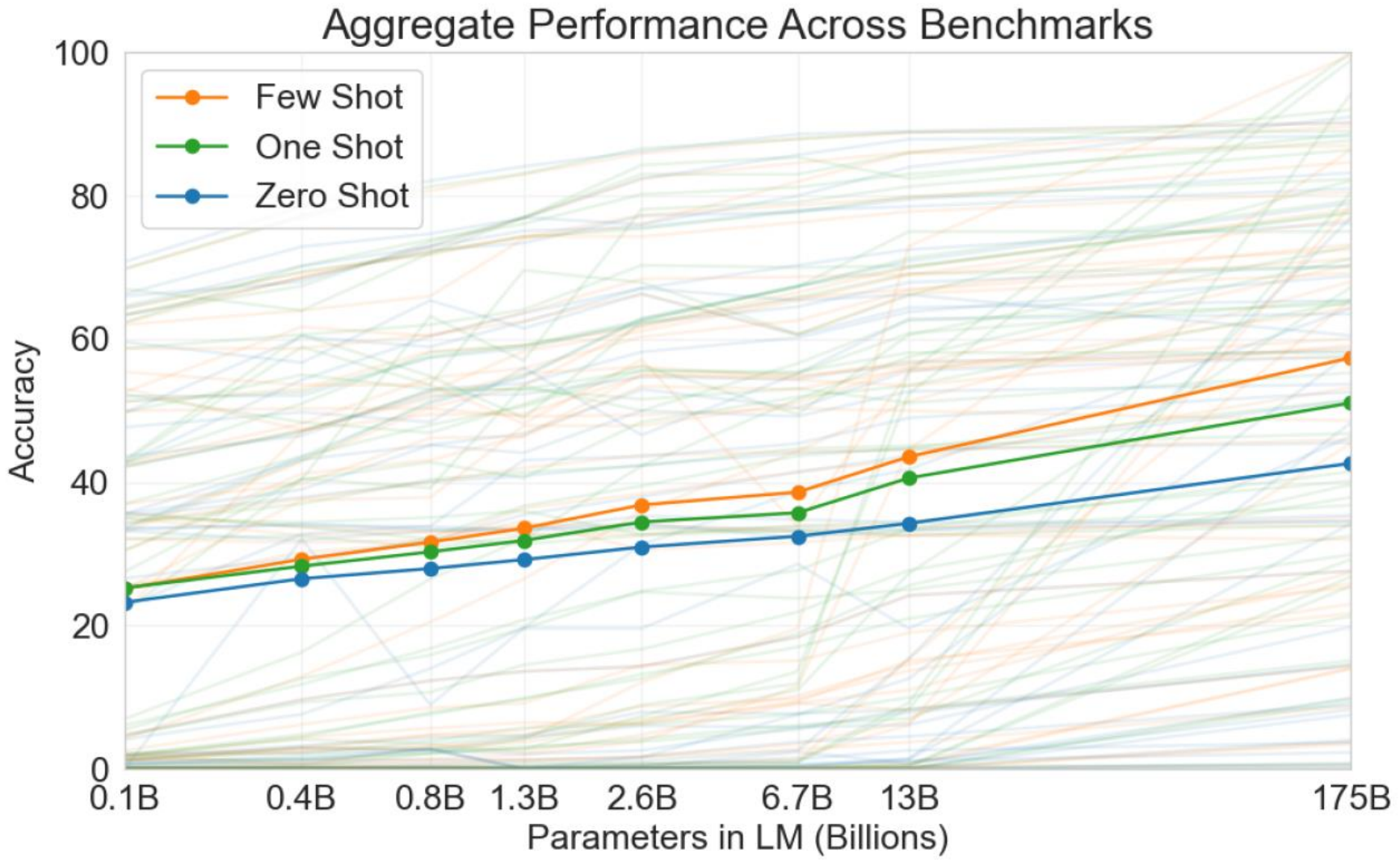
Q: What areas did Beyoncé compete in when she was growing up?
A: singing and dancing

Q: When did Beyoncé release Dangerously in Love?
A: 2003

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Q: When did Beyoncé start becoming popular?
A:

# Zero vs. Few-Shot Prompting



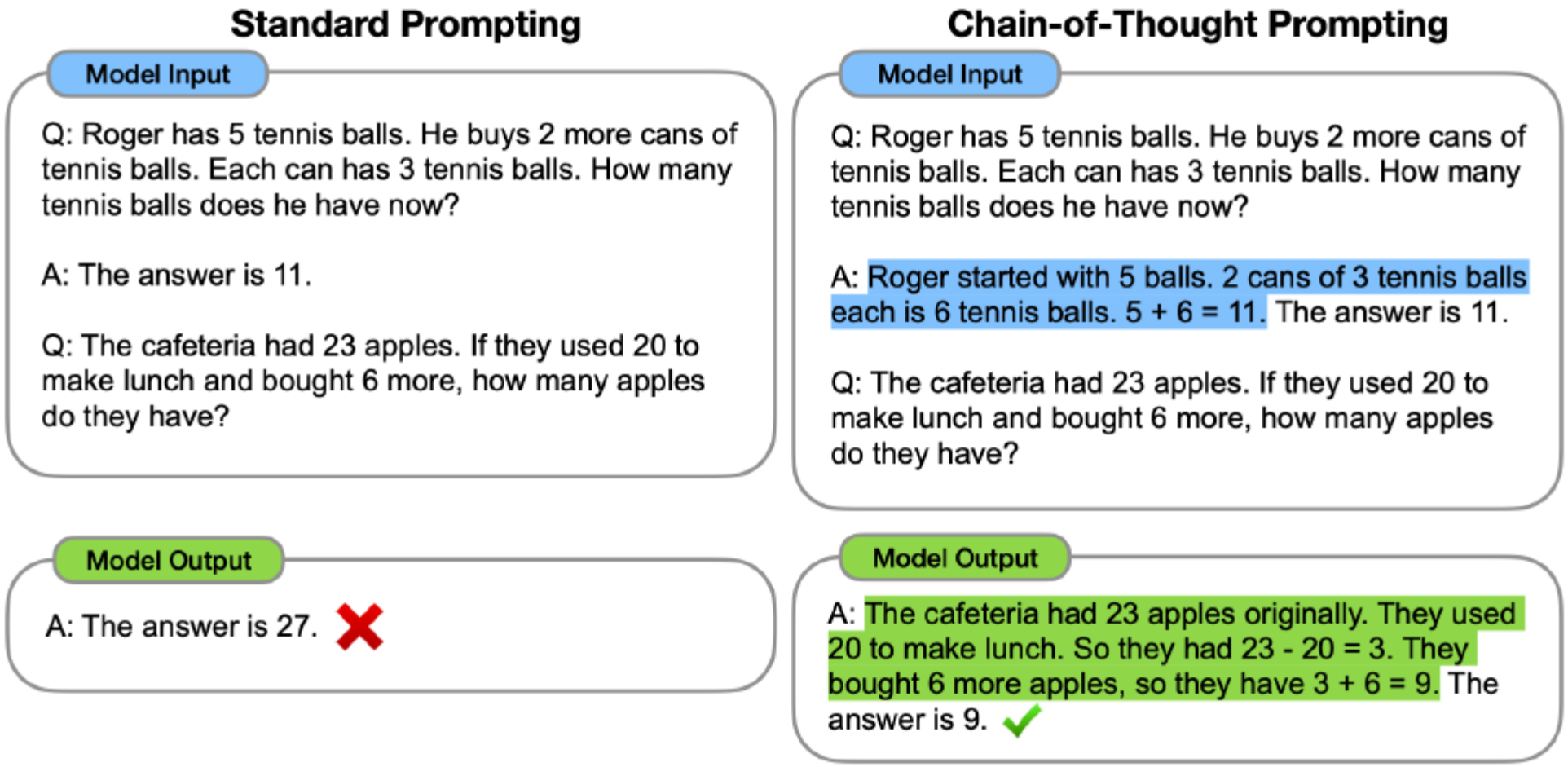



Results of GPT-3 from Brown, et al. (2020). "Language Models are Few-Shot Learners". NeurIPS. See figure 1.3.

# Zero vs. Few-Shot Prompting

| Prompting | Zero-Shot | Few-Shot |
|---|---|---|
| Pros | • No examples needed.<br>• Works for simple queries, and common knowledge tasks.<br>• Requires less effort than few shot prompting.<br>• Faster inference than few shot. | • Works for tasks requiring formatting or structure.<br>• Better for when a model struggles with zero-shot.<br>• Improved performance over zero-shot.<br>• Helps the model understand the task better. |
| Cons | • Can lead to incorrect results.<br>• May require multiple attempts to refine the prompt.<br>• Performance can be lower than few-shot prompting. | • A few examples needed.<br>• The demonstrations provided can impact model performance.<br>• It might still fail, and there is no warranty of good results. |

# Chain-of-Thought (CoT) Prompting




CoT improves LLM performance on tasks requiring reasoning by breaking them into steps—no gradient updates needed.



Wei, et al. ``Chain-of-Thought Prompting Elicits Reasoning in Large Language Models''. NeurIPS, 2022.

# In-Context Learning (ICL)

The term in-context learning (ICL) was first proposed by Brown et al. (2020) as part of their GPT3 system.

ICL means LLMs learning to do new tasks not explicitly trained for, better predict tokens at inference time, without any gradient-based updates to the model's parameters.

Benefits of ICL:

- No need for finetuning
- Flexible adaptation to different tasks
- Improved performance over zero-shot prompting
- Cost effective due to not retraining required



Whilst few-shot prompting is a technique, ICL is the ability that makes few-shot & CoT prompting work.

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# Potential Harms of LLMs

- LLMs are prone to saying things that are not true or correct, a term referred to as **hallucinations** (*"made-up responses"* without factual grounds). More on this issue next week!

- Although LLMs are finetuned for generating acceptable outputs, they could still generate **toxic language** (text that is harmful, offensive, or abusive such as discrimation, insults, misleading.)

- LLMs can be used by people with bad intentions for generating text involving **misinformation**, **phishing**, or other harmful activities.

# Today

- **Vector semantics**: words as vectors (count-based) to represent word meaning in NLP.
- **Word/Sentence Embeddings**: static, contextual (dense vectors)
- **Text similarity**: high application in NLP tasks.
- **Embedding-based classification**: LR, MLPs, among others.

# What's Next?

- Readings:
  - Jurafsky and Martin. Speech and Language Processing. Chapter 7, 2026. [recommended]
  - Zhao, et al. (2025). "A Survey of Large Language Models". CoRR.

- Workshop **tomorrow**: Prompting LLMs and LLMs as text classifiers.

- Next lecture: Question Answering & Info. Retrieval